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MINUTO FINAL REPORT

Measuring Solar Irradiance



IMT Atlantique
Bretagne-Pays de la Loire
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1 Introduction

Project context: Measuring solar irradiance with a thermal method

The solar irradiance received at ground level is a fundamental yet highly variable quantity. While the solar constant outside the atmosphere is $1360.8 \pm 0.5 \text{ W m}^{-2}$, it drops to about 1.2 kW m^{-2} at sea level under the most favourable conditions.

In Nantes, this value fluctuates drastically with the weather and the seasons: from 1016 W m^{-2} under clear skies in June to less than 160 W m^{-2} in winter.

Understanding and measuring this resource is crucial for sizing energy installations and for understanding heat transfer.

Rather than using photovoltaic panels, whose calibration paradoxically requires the irradiance to be known already, the MINUTO project relies on the design of a thermal sensor.

This method uses a black body (a brass cube) as an absorber, whose measured temperature rise makes it possible to derive the received solar flux indirectly.

Objectives: Designing a precision instrument

The technical mission is built around three major requirements defined by the specifications and by our group's ambitions:

Precision and Performance:

- Build an electronic thermometer able to detect temperature variations of $\pm 0.03 \text{ }^{\circ}\text{C}$ (our prototype achieved a precision of $0.018 \text{ }^{\circ}\text{C}$).
- Measure the specific heat capacity (C_T) of the metal block to within $\pm 3\%$.
- Determine the solar irradiance with a relative uncertainty below 10%.

Reproducibility and Robustness:

- The instrument must be ergonomic, portable and reliable for long in-situ measurements (at least one hour).
- Acquisition must be automated with an Arduino microcontroller to ensure data traceability.

Eco-design (cross-cutting objective):

- Use inexpensive, reusable equipment.
- Include a critical reflection on environmental impact (embodied energy, material life cycle) in order to propose a sustainable iteration of the sensor, notably by replacing petroleum-based materials with bio-based alternatives and by studying energy autonomy (*Solar-Powered*).

Team presentation

This project was carried out by our team: Omar AMARA, Kellian BAILLEUL, Pierre BERNARD and Lucille AVIGNON.

To carry out the design of our precision thermometer and the analysis of its impact, we set up an organization based on complementary skills and good coordination.

We defined specific areas of responsibility:

- Circuit lead: Pierre BERNARD
- Code lead: Omar AMARA
- Theory lead: Kellian BAILLEUL
- Technical lead: Lucille AVIGNON



Figure 1: We are group 1!

Details of our working methodology, how we handled working remotely, and the review of our collaboration are given in Section 5.

2 Thermometer description

Temperature is measured by an electronic acquisition chain designed to maximize sensitivity around the useful range. The device is based on an NTC thermistor placed in a Wheatstone bridge, followed by an amplification stage and digitization.

2.1 Signal conditioning

To exploit the resistance variation of the thermistor (Steinhart-Hart model), we use a Wheatstone bridge, which measures a relative potential difference and thus cancels the offset. This voltage is then amplified by an integrated operational amplifier (op-amp).

Based on our measurements and the circuit design:

- The gain of the amplifier stage is approximately 4.5.
- The saturation voltage of the op-amp was measured at 3.70 V.
- To ensure measurement stability, the bridge reference voltage U_{ref} is not taken from the general power supply but is imposed and regulated directly by the Arduino.

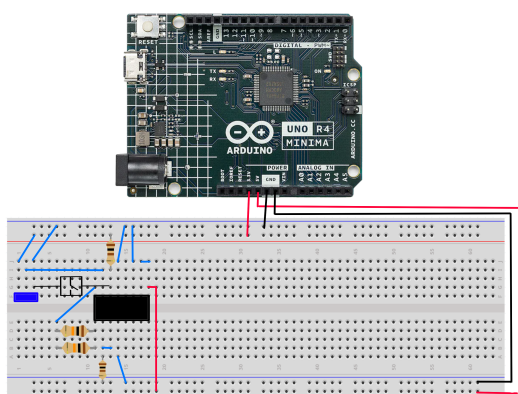


Figure 2: Circuit schematic (bridge and amplification).

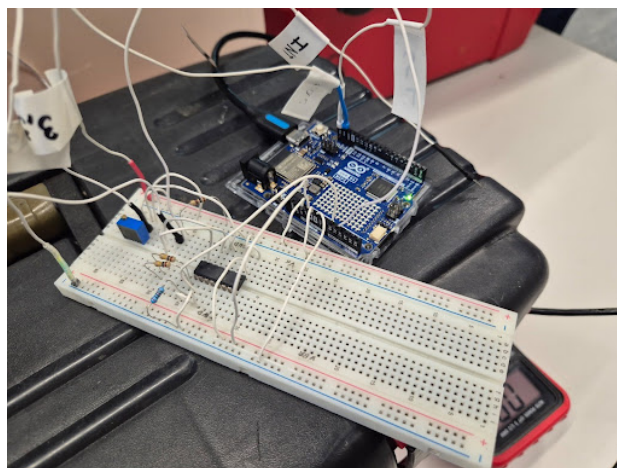


Figure 3: Breadboard implementation.

2.2 Design rationale

Our thermometer is based on a Wheatstone bridge circuit because, after analysis, we concluded that a simple voltage divider would not be sufficient for the required precision of 0.03°C . An op-amp was necessary to amplify the signal, since the analog reading was on 11 bits. However, the voltage must not exceed 4.7V, the maximum voltage of the analog inputs. We therefore chose the Wheatstone bridge approach with a gain of approximately 4.5. It is important to note that our op-amp has a saturation voltage of 3.70V, so our data range lies between 0.00V and 3.70V. Moreover, the voltage across the thermistor increases when the temperature decreases. Thus, since the op-amp input voltage is $U_{ref} - U_{th}$, we set U_{ref} equal to the voltage across the thermistor at $T = 2.30^{\circ}\text{C}$. This parameter allowed us to read temperatures over a range where $T_{min} = 2.30^{\circ}\text{C}$, which comfortably meets the specification requiring $T_{min} \leq 5.0^{\circ}\text{C}$.

After the calibration of our thermometer, which went very well, we observed during some of the measurements presented in the next section inconsistent fluctuations of our U_{ref} voltage, which

corrupted our acquisitions. We therefore changed the way U_{ref} is generated, outputting a fixed voltage directly from the Arduino board by adding the following lines to our voltage-reading code:

Code lines configuring the analog output (Arduino)

```
analogWriteResolution(12);  
analogWrite(A0, 1828);
```

2.3 Performance and Calibration

Acquisition is performed on 11 bits (after a bit shift of a 12-bit reading), giving 2047 quantization levels (N_q). Calibration was carried out by comparison with a reference probe (TMP 117). We thus record a voltage U_g as a function of time and a temperature as a function of time. The accuracy of the TMP is given as $u(T) = 0.1\text{ }^\circ\text{C}$. The uncertainty on U_g is $u(U_g) = \frac{V_{ref}}{N_q} = \frac{4.86}{2047} \approx 2.37\text{ mV}$. We then ran a Monte Carlo simulation ($n = 1000$) with the following linear regression: $U_g = a \cdot T + b$, from which we obtained the calibration coefficients a and b . It is important to note that this simulation was performed after removing the points corresponding to the jumps in U_g (see FIGURE 4). We obtained:

$$a = (0.1257 \pm 0.0001)\text{ V} \cdot \text{C}^{-1}$$

$$b = (-0.2890 \pm 0.0017)\text{ V}$$

The results obtained (see Figure 6) show:

- An effective measurement range from 3.85°C to 30.09°C .
- An experimental precision of $\pm 0.018^\circ\text{C}$.

This precision is below the 0.03°C limit set by the specifications, thus validating the design for its intended metrological use.

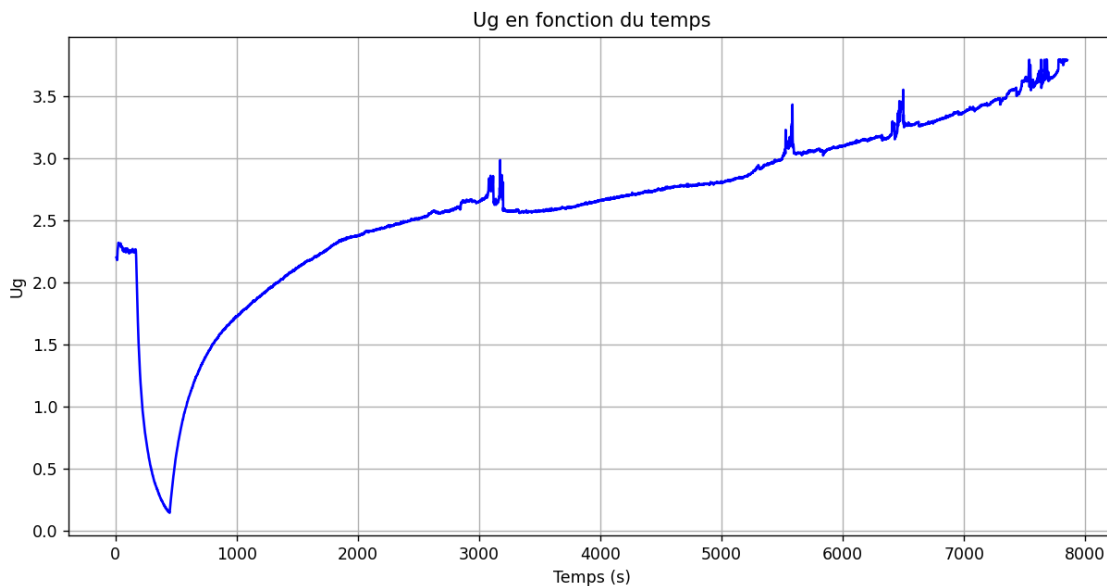


Figure 4: Evolution of U_g over time.

3. Measuring the heat capacity C_T

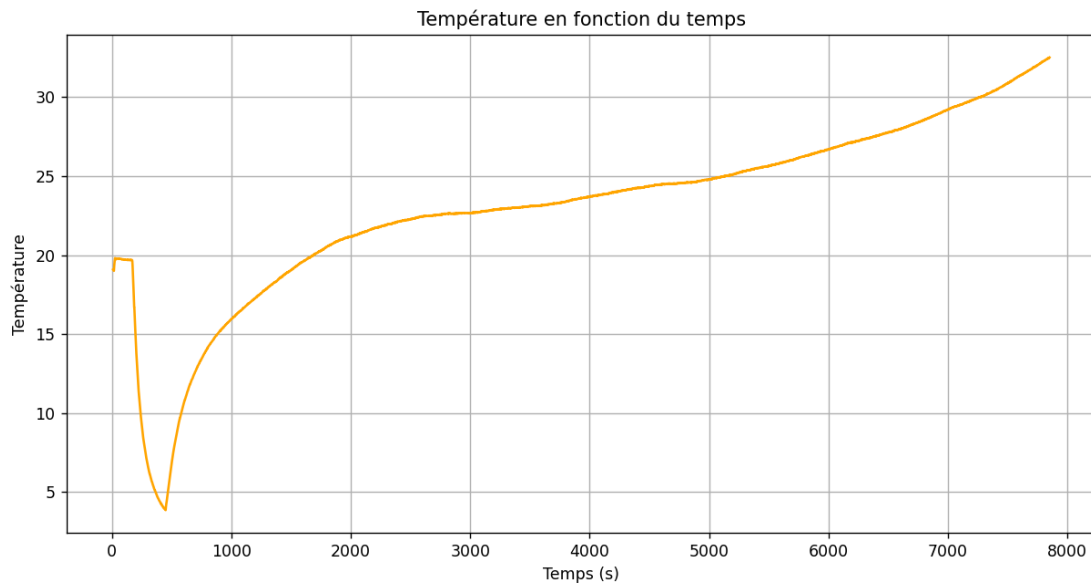


Figure 5: Evolution of T over time (TMP 117).

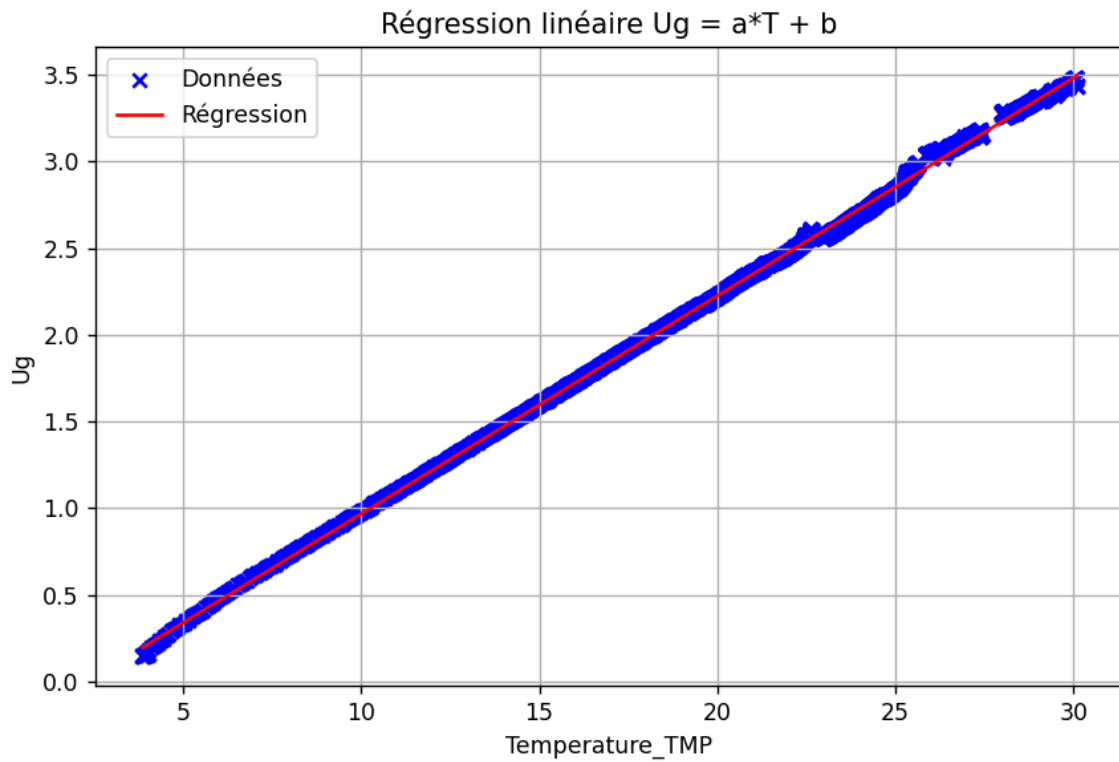


Figure 6: Regression plotted for $\bar{a} = 0.1257 \text{ V} \cdot \text{C}^{-1}$ and $\bar{b} = -0.2890 \text{ V}$.

The residual plots (deviations of the measured values from the mean regression with $\bar{a}$ and $\bar{b}$) and the distributions of the values of a and b from the Monte Carlo simulation can also be seen in Figure 15 and Figure 14.

3 Measuring the heat capacity C_T

The overall heat capacity of the system (C_T) is determined by an active calorimetric method (Joule heating). This constant is essential for subsequently deriving the solar flux.

3.1 Experimental protocol: Measurement in a cold environment

Since our sensor saturates at 31.76°C , it was impossible to achieve significant heating at room temperature (20°C) without exceeding the sensor's limit. We therefore adopted the following protocol:

1. Place the material with its insulating block in the cold environment for 1 hour to stabilize the setup.
2. Insert the cube into its insulating block, then leave the whole assembly for an additional 15 minutes for a second stabilization phase.
3. Start data acquisition.
4. Once acquisition has started, connect the heating wire.

The theoretical model of the temperature evolution is given by:

$$T(t) = T_{ext} + \frac{P}{\lambda_{eff}} \left(1 - e^{-\frac{\lambda_{eff} t}{m C_T}} \right)$$

3.2 Results and Uncertainties (Monte Carlo)

The data were analysed using a Monte Carlo simulation ($N = 100\,000$ draws) to propagate the uncertainties on the voltage (U_{wire}), the resistance and the time.

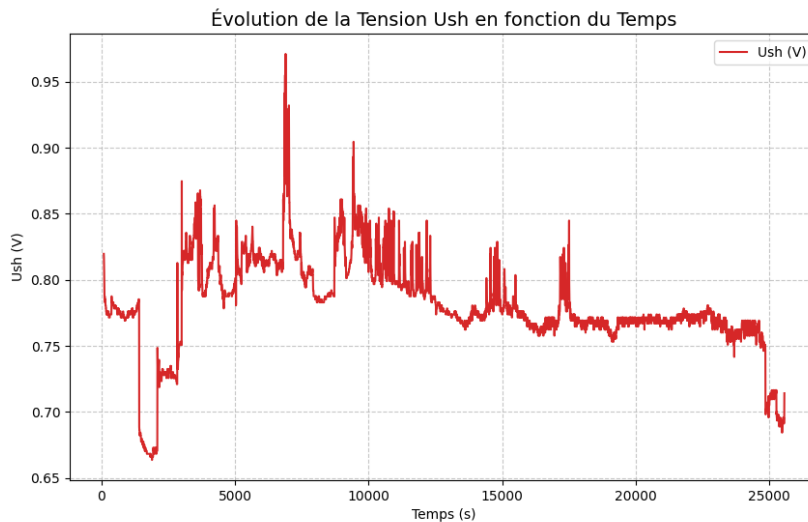


Figure 7: Stability of the voltage across the heating wire during the experiment.

The resulting distribution histograms give the mean value and standard deviation of the heat capacity:

3. Measuring the heat capacity C_T

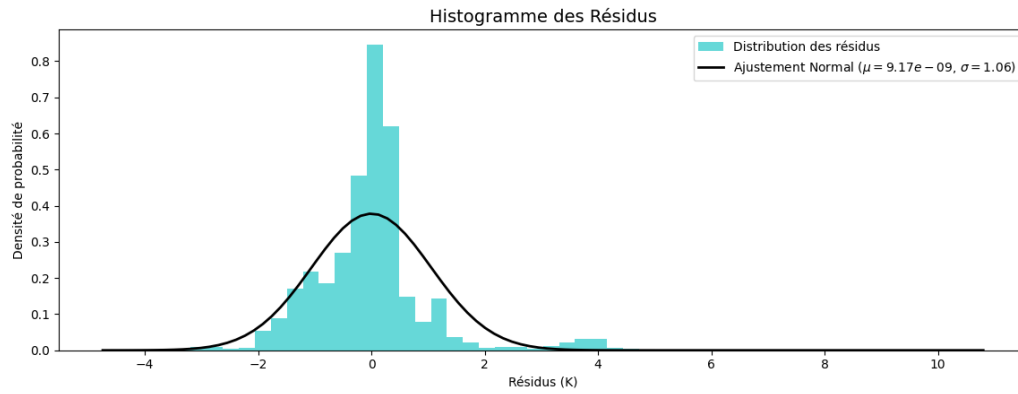


Figure 8: Distribution of C_T obtained by the Monte Carlo method.

We finally obtain:

$$C_T = 378.48 \pm 5.37 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$$

i.e. a relative uncertainty of 1.3%, which meets the 3% target.

This value is highly consistent with the theoretical specific heat capacity of **brass** ($377 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$), confirming the nature of the material the sensor block is made of.

4 Solar data analysis

The final objective is to determine the solar irradiance $F_0(t)$ from the thermal behaviour of the sensor exposed to the sun.

4.1 Thermal model

The heat balance of the sensor, treated as a black body, accounts for energy storage, radiative and convective losses, and the solar input:

$$mC_T \frac{dT}{dt} = -\sigma S_0(T^4 - T_{ext}^4) - hS_0(T - T_{ext}) + F_0(t)S_e$$

where $S_e = S_0 \cos(\theta)$ is the effective surface exposed to the sun. Before computing the flux, we determined the effective thermal conductance λ_{eff} of the insulated system through a regression on the relaxation phase. We find:

$$\lambda_{eff} = (38.98 \pm 3.41) \times 10^{-3} \text{ W} \cdot \text{K}^{-1}$$

4.2 Outdoor protocol

1. **Acclimatization:** the brass cube and its block were placed outdoors, in the shade, for 1 hour.
2. **Assembly and sealing:** after acclimatization, the cube was inserted into the block, then the setup was sealed with tape and a glass pane.
3. **Initial positioning:** the setup was placed on its stand, with the opposite face oriented towards the sun.
4. **Start and exposure:** data acquisition was started, then the cube was exposed directly to the sun.
5. **Observation:** the setup was left to run on its own for about 1 hour, under constant cloudless sunshine, without rotating the device.



Figure 9: Experimental setup: (left) experiment 1, (right) experiment 2

4.3 In-situ measurements and results

Two measurement campaigns were carried out in Le Mans under real conditions. For each session, the sensor was acclimatized in the shade before being exposed.

Campaign 1: Friday, 26 December

The sky was clear. Analysis of the data with the inverse model gives a mean flux of:

$$F_0 = 649.99 \pm 31.25 \text{ W} \cdot \text{m}^{-2}$$

Campaign 2: Sunday, 28 December

A second measurement was carried out to check repeatability and consistency with the variable weather conditions.

$$F_0 = 886.87 \pm 45.46 \text{ W} \cdot \text{m}^{-2}$$

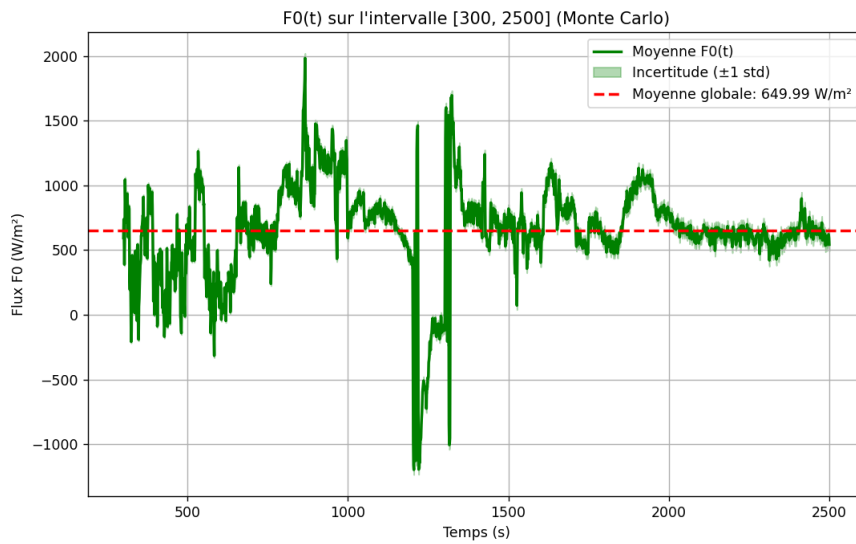


Figure 10: Solar flux plot — 26 December

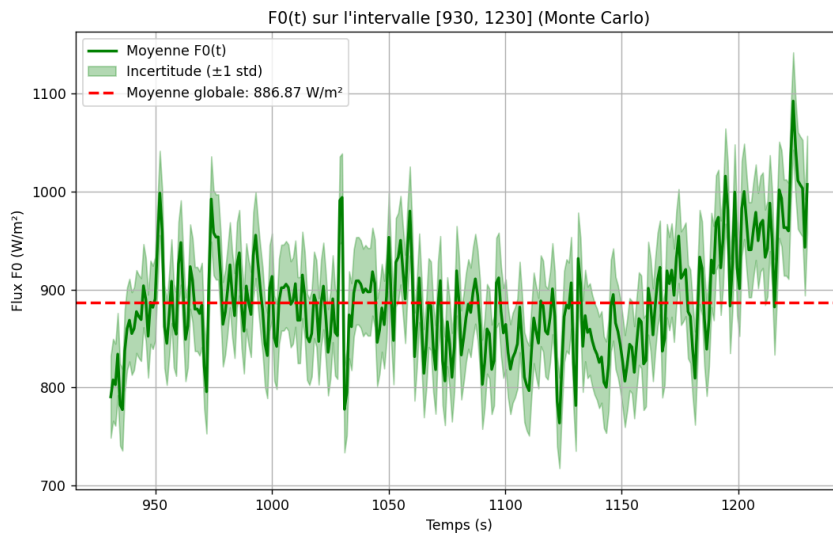


Figure 11: Solar flux plot — 28 December

4.4 Comparison of results with the IMT Atlantique sensors in Nantes

To compare the experimentally measured solar fluxes with reference data, they were compared with the IMT Atlantique sensors in Nantes. For each campaign, data were taken only over the time ranges corresponding to the effective analysis phase of the measurements, once the thermal regime had stabilized.

From these intervals, the sun's position at the given time was determined, making it possible to derive the normal incident irradiance from the irradiance measured on the panels. The results obtained for each campaign are presented below.

4.4.0.1 Campaign of Friday, 26 December Acquisition started at 11:50. The usable data range corresponds to the interval between 2000 s and 2500 s, i.e. an analysis carried out between 12:23 and 12:30. The mean solar flux estimated by the inverse model is:

$$F_0 = 650 \pm 31.25 \text{ W} \cdot \text{m}^{-2}$$

At that time, the solar elevation angle was 18.67° . The irradiance measured on the IMT Atlantique panels is $267 \text{ W} \cdot \text{m}^{-2}$, which yields a calculated normal irradiance of:

$$E_{\text{norm}} = 834 \text{ W} \cdot \text{m}^{-2}$$

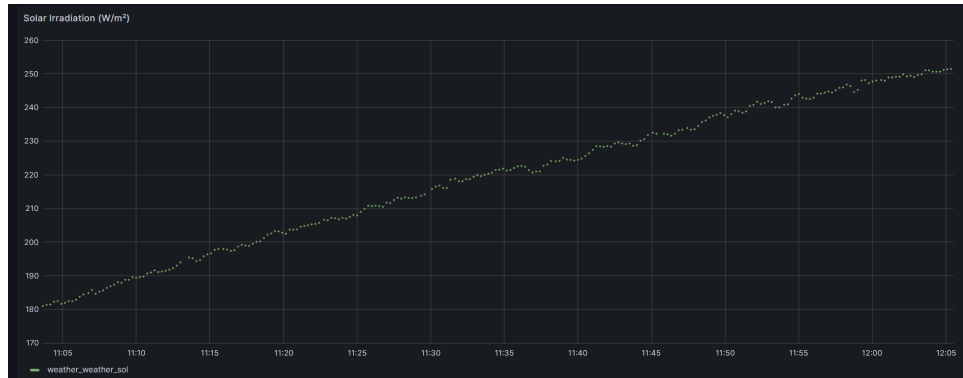


Figure 12: Irradiance measurement — Friday, 26 December

4.4.0.2 Campaign of Sunday, 28 December The second acquisition started at 12:32. The data range used for the analysis extends from 900 s to 1250 s, corresponding to the time interval between 12:47 and 12:53. The estimated solar flux is then:

$$F_0 = 887 \pm 45.5 \text{ W} \cdot \text{m}^{-2}$$

The solar elevation angle at the time of analysis was 19.35° . The irradiance measured on the panels is $150 \text{ W} \cdot \text{m}^{-2}$, which yields a calculated normal irradiance of:

$$E_{\text{norm}} = 452 \text{ W} \cdot \text{m}^{-2}$$

4. Solar data analysis

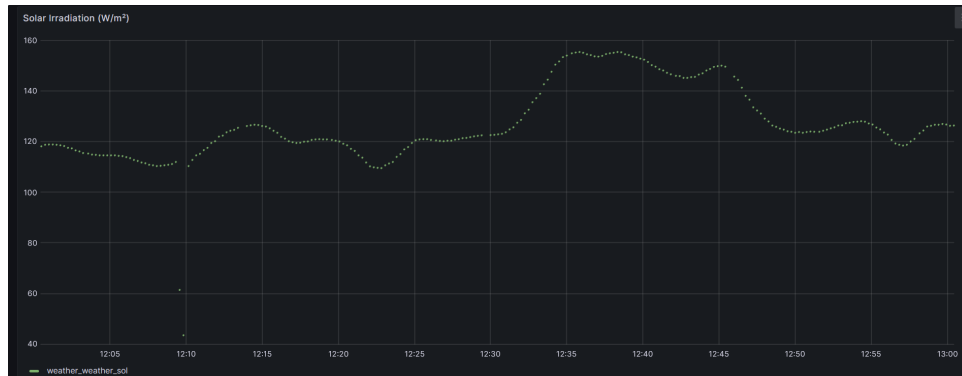


Figure 13: Irradiance measurement — Sunday, 28 December

4.5 Discussion

The comparisons between the solar fluxes reconstructed from the experimental measurements and the data from the IMT Atlantique reference weather sensors in Nantes show satisfactory overall consistency. The flux values obtained with the inverse model are of the same order of magnitude as the normal irradiances calculated from the panel measurements, which confirms the relevance of the method used.

Overall, the expanded uncertainty associated with the measurements remains below 10 %, which satisfies the requirements of the specifications. The rapid variations observed in the computed flux are consistent with the real atmospheric conditions and confirm the device's sensitivity to changes in irradiance and to heat exchange with the environment. These results validate the experimental approach while highlighting the limitations due to uncontrolled outdoor conditions.

5 Work organization and team functioning

From the start of the project, our team set up a rigorous organization based on complementary skills and continuous communication. To ensure motivated involvement and optimal efficiency, we chose to assign tasks according to each member's interests and areas of expertise.

This approach takes the form of four main roles: circuit lead, technical lead, code lead and theory lead. It is important to stress that this distribution does not create any hierarchy; we deliberately maintain a horizontal organization, ensuring that everyone participates equally in strategic decisions and in the overall monitoring of progress.

5.1 Roles defined by expertise and mutual learning

The Circuit Lead is responsible for designing and drawing up the complete electronic schematic. Their responsibilities include selecting components, preparing wiring plans and checking the internal consistency of the circuit, thus preparing the practical implementation phase. In close collaboration, the Technical Lead handles the practical implementation: soldering, fitting heat-shrink tubing, final assembly of the components and all experimental checks.

The equipment required for this work is contained in the Minuto kit. The kit circulates regularly between team members (except for the theory lead, who does not live on campus) so that everyone can work on the project in the evenings and at weekends.

In parallel, the Code Lead is dedicated to developing and optimizing the program associated with the device. Their role consists of debugging, proposing continuous improvements and keeping the GitHub repository up to date, ensuring traceability and software follow-up for the whole team. Finally, the Theory Lead ensures the scientific consistency of the project by checking the accuracy of the physical laws used and clarifying complex concepts for the team when necessary. Writing on Overleaf is done collectively and is not assigned to a specific role.

It should be noted that, while these roles structure the work, they do not imply rigid specialization. We encourage interest in other areas and mutual learning, so that the whole team acquires cross-disciplinary knowledge.

5.2 Coordination and flexibility

Beyond these specific functions, several responsibilities are shared collectively.

Communication is maintained continuously through a WhatsApp group, our main channel for quickly sharing unexpected issues, progress or difficulties. In addition, weekly meetings form the backbone of our organization. They mainly take place during the class hours dedicated to the project and, when needed, we book a dedicated room through the Caræ system for structured work sessions. These meetings allow us to structure the work, take precise stock of progress and, if necessary, refocus our priorities.

The secretary role, held on a rotating basis, ensures rigorous traceability of the decisions taken during these sessions. More broadly, overall coordination, task allocation and progress assessment are also carried out collectively, strengthening each member's involvement and sense of responsibility.

5.3 Review and outlook

So far, the initial organization has proven highly effective. The distribution of roles is relevant, the working atmosphere remains positive and each member progresses at a pace suited to their constraints.

5. Work organization and team functioning

We nonetheless remain flexible and regularly adapt how we work, notably by structuring our meetings more and adding extra work sessions to solve technical problems. Gradually, our shared documentation, initially loosely structured, is being consolidated, making GitHub and Overleaf central tools for pooling our work.

To further improve our efficiency, we have already identified several areas for improvement. We will establish a more detailed schedule with clearly defined intermediate milestones, to better anticipate critical stages. Meeting efficiency will be improved by alternately appointing a facilitator and a secretary, in order to systematize decision-making and guarantee impeccable traceability.

In addition, we will structure the shared documentation on GitHub and Overleaf from the outset and anticipate equipment needs and room bookings on Caræ more rigorously to minimize logistical constraints. Finally, we will organize dedicated knowledge-sharing sessions earlier and more regularly.

This ensures that every member fully masters the key concepts required for an overall understanding of the project.

6 Reflection on the environmental impact of the project

6.1 The invisible footprint: embodied energy and manufacturing

The environmental analysis of our precision thermometer reveals a reality common to technological objects: most of the impact is “invisible”. What is known as embodied energy (the energy consumed to extract, process and transport materials) accounts for the predominant share of our carbon footprint, estimated at between 3 and 3.5 kg CO₂e. By comparison, this footprint corresponds to less than 20% of the emissions of a one-hour journey in a combustion-engine car (about 20 kg CO₂e).

For a single prototype, this figure may seem modest, but it is still equivalent to manufacturing about ten cotton T-shirts or to using an LED bulb for several years.

However, the manufacturing of the active components, such as the Arduino microcontroller or the instrumentation amplifier, is the critical point, accounting for more than 70% of the total impact. Although tiny, these chips require lithography processes in extremely energy-intensive clean rooms, where air conditioning and the purity of the gases used carry an environmental cost that is disproportionate to the mass of the finished product.

According to ADEME data, the carbon intensity of electronics production is among the highest per kilogram of material, which justifies our choice to favour reusable components over dedicated single-use printed circuit boards (PCBs).

6.2 Material substitution: towards bio-based alternatives

A major focus of our reflection is thermal insulation. We currently use an expanded polystyrene box, a petrochemical material whose recycling is complex and unprofitable. Its environmental persistence in the form of microplastics is a major drawback.

Research on innovative materials suggests promising alternatives that we could explore in a future iteration. For example, mycomaterials (a foam made from mushroom roots, i.e. mycelium) can achieve thermal insulation performance of 0.018°C, comparable to polystyrene, while being fully compostable and carbon-neutral during their growth, ensuring the biodegradability of the system.

Another option would be expanded cork, which offers excellent thermal inertia while being a natural carbon sink.

These substitutions would drastically reduce the environmental persistence of the sensor at the end of its life.

6.3 Circularity and recovery of metals

Conversely, the brass cube used as the thermal absorber, the central element of our device, perfectly illustrates the principles of the circular economy.

Although the initial extraction of copper and zinc is polluting, brass has a recyclability rate close to 100%; it is said to be infinitely recyclable without loss of properties. Drawing on industrial models such as that of the company Boutté, we note that recycling brass consumes 90% less energy than its primary production.

This intrinsic durability turns an initial environmental cost into material capital that can be reused almost indefinitely, offsetting its carbon weight of 0.7 kg CO₂e in the long run.

Regarding the passive components (resistors, thermistor), although their individual recycling is difficult due to their small size, their impact remains marginal compared with the connectors. We have nonetheless become aware that the use of tin for soldering, even in small quantities, remains

problematic because of mining conditions. A possible improvement would be to use mechanical connectors or solders based on lower-impact alloys.

6.4 Low consumption during use and future prospects

During acquisition, the system is remarkably frugal, consuming only a few hundred milliwatts. However, a systemic view requires including the energy consumed by the control computer, which highlights a paradox: the measuring device itself is exemplary in its frugality, but it depends on a computer for data processing.

Used for programming and data processing (notably for the Monte Carlo simulations), the computer accounts for a much larger share of consumption than the sensor itself.

To reduce this impact in the future and make our project truly “low-power”, several eco-design options can be considered.

The next step would be to migrate to a standalone data storage system on an SD card, powered by a small photovoltaic cell.

This “Solar-Powered” approach would be fully in line with the primary function of our instrument: measuring solar irradiance. By removing the permanent link to the computer, we would divide the operating footprint by a factor of ten, turning a laboratory tool into an autonomous and sustainable field instrument.

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7 Appendices

7.1 Residuals and distributions for the Monte Carlo calibration simulation

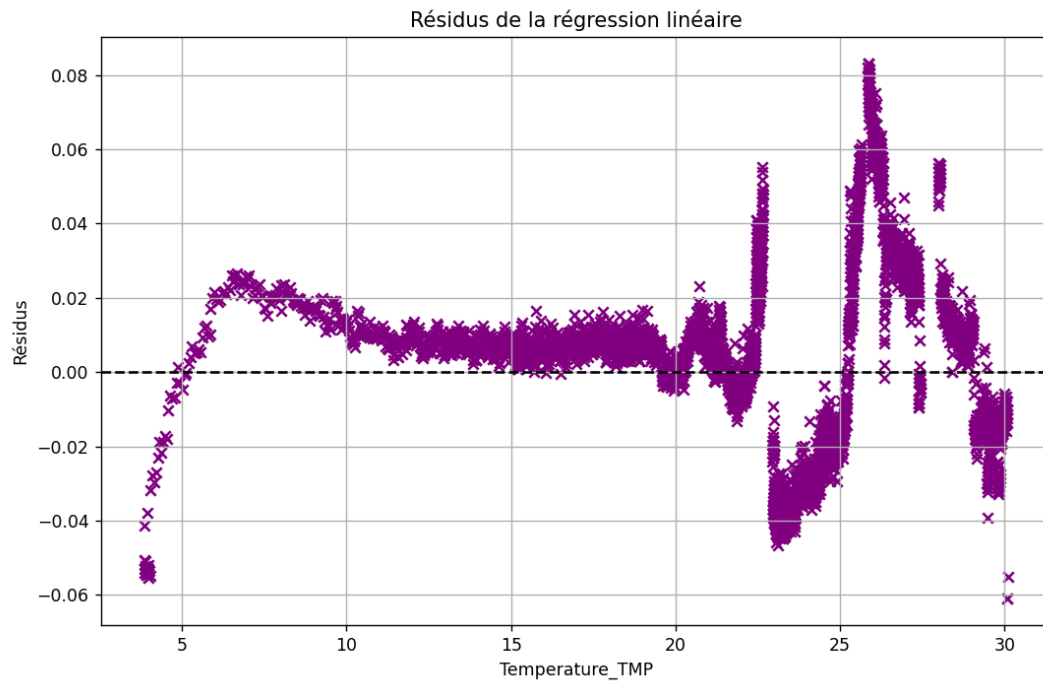


Figure 14: Residual deviations from the calibration simulation (final T computed with $\bar{a}$ and $\bar{b}$ compared with T from the TMP 117)

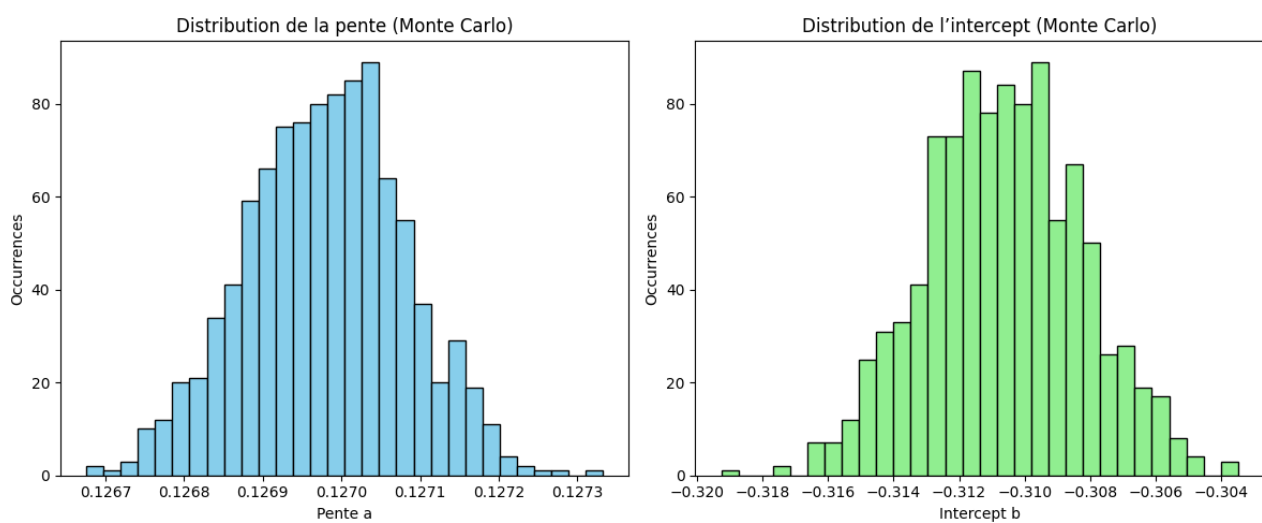


Figure 15: Distributions from the calibration simulation